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Student Number



Barker
College

2023

**YEAR 12 ASSESSMENT BLOCK
SEMESTER 1**

Mathematics Advanced

Staff Involved:

- | | | |
|----------------------|------------------------|----------------------|
| • DXC - Mr. Chua | • LMD - Mrs. De Gorter | |
| • JAI - Ms. Iles | • DZB - Mr. Batchen | |
| • LAK - Mrs. Kalnins | • ARM - Mr. Mallam | • RJW - Mr. Williams |
| • PDJ - Ms. Jouliany | • ARP - Mr. Perkins | • ALY - Mrs. Young |

FRIDAY 31 MARCH 2023

TOPICS COVERED:

Year 11 Content	Sequences & Series
Series & Finance	Graphs & Equations
Curve Sketching using the Derivative	Integration
Trigonometric functions	

150 copies

General

Instructions:

- Write your Student Number at the top of every page
- Working time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations

Total marks:

80

Section I - 8 marks (pages 2 - 5)

- Attempt Questions 1 - 8
- Allow about 15 minutes for this section

Section II - 72 marks (pages 7 - 27)

- Attempt Questions 9 - 17
- Allow about 1 hour and 45 minutes for this section

Section I

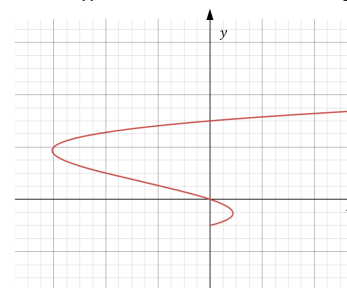
8 marks

Attempt Questions 1 – 8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 8.

1. What type of relation is shown in the graph below?



- A. One-to-one
- B. One-to-many
- C. Many-to-one
- D. Many-to-many

2. The wheel below displays the numbers 1-20 exactly once.



If the wheel is spun 160 times, how many times would you expect a number less than 6 to be obtained?

- A. 24
- B. 32
- C. 40
- D. 48

3. What is the derivative of $\frac{x}{\cos x}$?

- A. $\frac{\cos x + x \sin x}{\cos^2 x}$
 B. $\frac{\cos x - x \sin x}{\cos^2 x}$
 C. $\frac{x \sin x - \cos x}{\cos^2 x}$
 D. $\frac{-x \sin x - \cos x}{\cos^2 x}$

4. A geometric sequence has first term 20 and common ratio -1.5 .

What is the value of T_5 ?

- A. 12.5
 B. $-151\frac{7}{8}$
 C. $101\frac{1}{4}$
 D. $-101\frac{1}{4}$

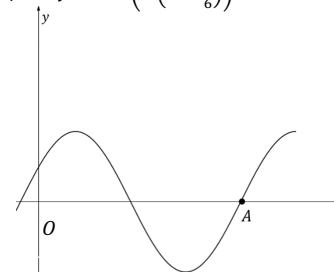
5. A discrete random variable X has the following probability distribution.

x	0	1	2	3
$P(X = x)$	a	$3a$	$5a$	$7a$

Which of the following is $E(X)$?

- A. $\frac{1}{16}$
 B. $\frac{17}{8}$
 C. $\frac{35}{16}$
 D. 1

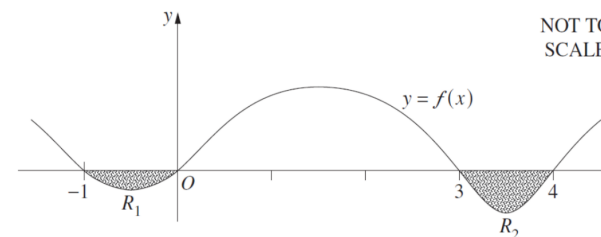
6. The graph of $y = \cos\left(2\left(x - \frac{\pi}{6}\right)\right)$ is shown below.



What are the co-ordinates of A ?

- A. $\left(\frac{5\pi}{12}, 0\right)$
 B. $\left(\frac{2\pi}{3}, 0\right)$
 C. $\left(\frac{11\pi}{12}, 0\right)$
 D. $\left(\frac{7\pi}{6}, 0\right)$

7. The diagram shows the graph of $y = f(x)$ with intercepts at $x = -1, 0, 3$ and 4 .



The area of the region shaded R_1 is 2 square units.

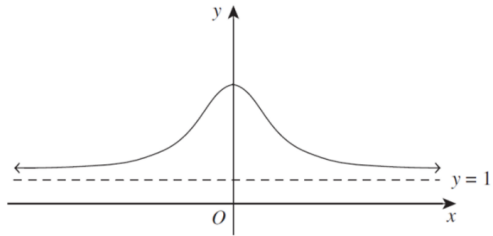
The area of the region shaded R_2 is 3 square units.

It is given that $\int_0^4 f(x) dx = 10$

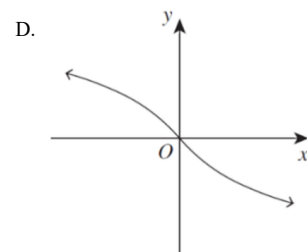
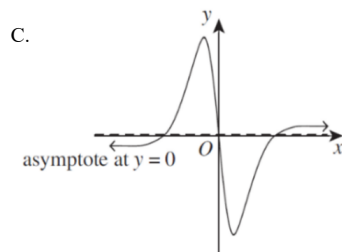
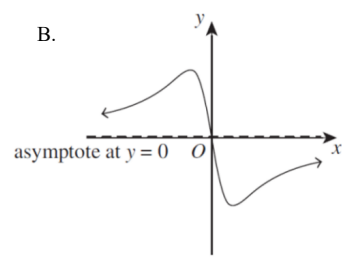
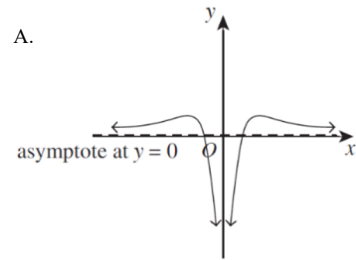
What is the value of $\int_{-1}^3 f(x) dx$

- A. 5
 B. 9
 C. 11
 D. 13

8. A sketch of the function $y = f(x)$ is shown below with a horizontal asymptote at $y = 1$.



Which of the following could be the sketch of $y = f'(x)$?



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End of Section I

Student Number

72 marks

Attempt Questions 9 – 17

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate space provided.

In Questions 9 – 17, your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (8 marks)

(a) Find the sum of the arithmetic series

3

$$9 + 16 + 23 + \dots + 2963$$

[illegible]

(b) Differentiate

$\alpha.$ $y = 3 + \sin 2x$

1

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.....

.....

β. $f(x) = e^x(x^2 + 1)$

1

[illegible]

(c) Find the exact gradient of the tangent to the curve $y = x \tan x$ at the point

on the curve where $x = \frac{\pi}{3}$.

3

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Student Number

Question 10 (9 marks)

(a) Find the following:

$\alpha. \int \frac{x^2 + 2x}{x} dx.$

1

$\beta. \int (3x - 4)^8 dx.$

1

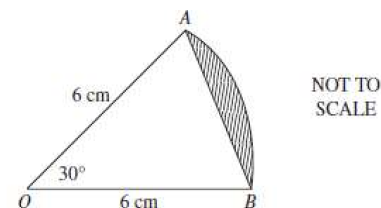
(b) (i) Sketch the function $y = \sqrt{4 - (x - 2)^2}$, clearly showing any x -intercepts.

3

(ii) Hence, exactly evaluate $\int_2^4 \sqrt{4 - (x - 2)^2} dx.$

1

(c) In the diagram OAB is a sector with center O and radius 6cm with $\angle AOB = 30^\circ$.



(i) Find the area of the triangle OAB .

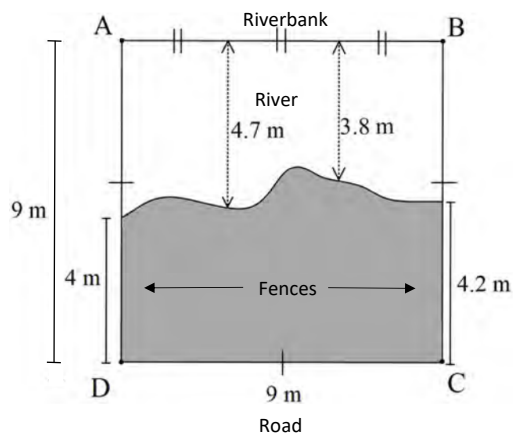
1

(ii) Find the exact value of the area of the shaded segment.

2

Student Number

Estimate the area of the paddock using the Trapezoidal rule with four function values.

[illegible]

1

[illegible]

2

[illegible]

2

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Student Number

(a) Given that $f(x) = 3x - 1$ and $g(x) = \frac{1}{2x}$ find:

1

1

3

A Cartesian coordinate system showing the graph of a parabola and a line. The parabola is labeled $y = x^2 - 5x$ and the line is labeled $y = x - 5$. The parabola opens upwards with its vertex in the fourth quadrant. The line has a positive slope. They intersect at two points: point A in the third quadrant and the point $(5, 0)$ on the x-axis.

4

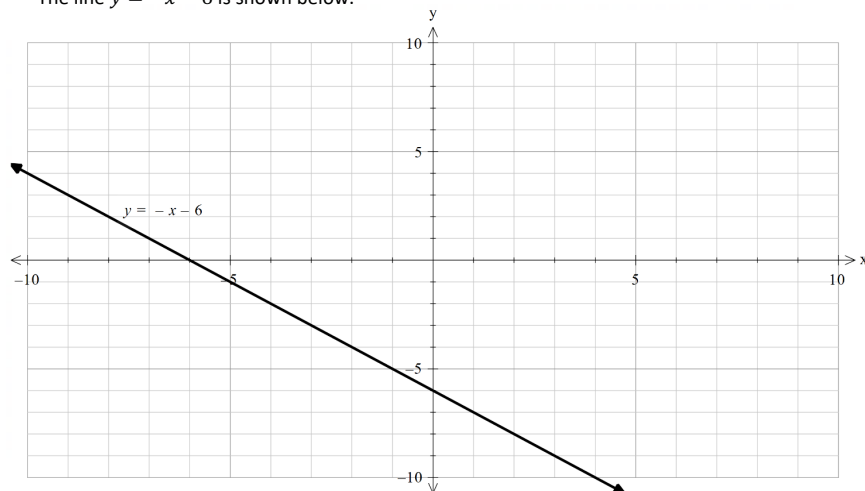
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Student Number

Question 13 (9 marks)

- (a) The curve $f(x) = x^2(x - 4)$ and the line $y = -x - 6$ intersect at the points $(-1, -5)$, $(2, -8)$ and $(3, -9)$.

The line $y = -x - 6$ is shown below.



- (i) Sketch the curve $f(x) = x^2(x - 4)$ on the graph above, showing the x and y intercepts, and the points of intersections with the line. **2**

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- (ii) Hence, solve $f(x) \leq -x - 6$, giving your answer in bracket interval notation. **2**

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- (b) Gracie plays football, a sport where a team can win, lose or draw.

Gracie is prone to injury and plays in 80% of her team's games.

When she plays, her team has a 73% chance of winning and a 7% chance of losing.

When she doesn't play her team has a 55% chance of winning and a 20% chance of losing.

- (i) By using a tree diagram, or otherwise, show that the probability that Gracie's team wins any given game is 69.4%. **2**

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- (ii) Given that Gracie's team doesn't lose in a particular match, what is the probability that Gracie played in that match? **3**

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Student Number

(a) The function $y = g(x)$ is transformed to the function $y = \frac{1}{2}g(3(x - 1)) + 4$.

Find the point $P(x, y)$ on $y = g(x)$, if it's image point on the **transformed** function is $P'(-6, 5)$.

3

- (b) Indie gets daily injections of a medicinal drug at the same time every day.

She gets d mg of the drug in each injection.

Each day the amount of the drug left in Indie's body from an injection decreases by 15%.

- (i) Show that the total amount of the drug (in mg) in Indie's body immediately after the n^{th} injection is given by:

3

$$A_n = \frac{20d(1 - 0.85^n)}{3}$$

- (ii) Hence, find the amount of the drug, d , in one of Indie's daily injections, correct to the nearest mg, if immediately after the 7th injection, there are 50 mg of the drug in Indie's body. 2

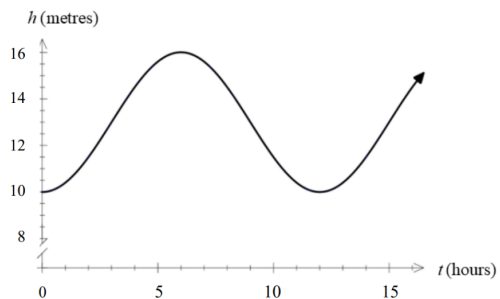
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Student Number

Question 15 (8 marks)

- (a) The rise and fall in sea level in Jervis Bay due to tides can be modelled by the cosine function: $h = A \cos(bt) + 13$, where h is the depth of the water in metres and t is hours since 8am.



At 8am it is low tide, and the bay is 10 m deep. At the next high tide at 2pm, the bay is 16m deep.

- (i) Find the values of A and b . **3**

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- (ii) A ship needs 11.5 m of water to sail and only sails in daylight.
Between exactly what times on the first day can the ship sail? **2**

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- (b) The sum of the first 10 terms of the following series is -110 :

3

$$\log_2 \left(\frac{1}{x} \right) + \log_2 \left(\frac{1}{x^2} \right) + \log_2 \left(\frac{1}{x^3} \right) + \dots$$

Find the value of x .

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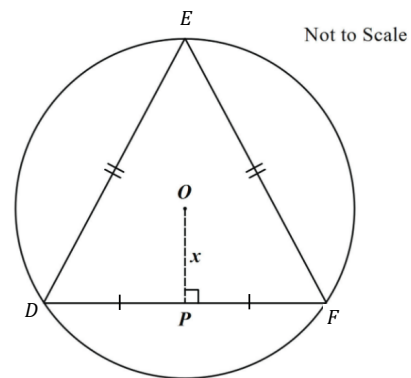
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Student Number

An isosceles triangle DEF , where $DE = EF$, is inscribed in a circle of radius 10 cm. $OP = x$ and OP bisects DF , such that $DF \perp OP$.



(ii) Show that the area, A , of $\triangle DEF$ is given by $A = (10 + x)\sqrt{100 - x^2}$. **2**

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- 25 -

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Student Number

Question 17 continued

By first showing that the derivative is:

$$\frac{dA}{dx} = \frac{100-10x-2x^2}{\sqrt{100-x^2}}$$

prove that the triangle with maximum area is equilateral.

5

End of Paper

Year 12

ADV MATHS TI EXAM SOL'NS 2023

Page 1/6

SECTION I: MULTI-CHOICE

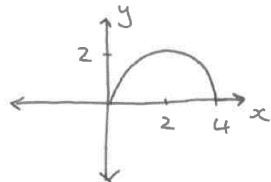
- ONE INPUT, MANY OUTPUTS. **[B]**
- $\frac{5}{20} \times 160 = 40$. **[C]**
- USE QUOTIENT RULE. **[A]**
- $T_5 = 20 \times (-1.5)^4$
 $= 101 \frac{1}{4}$. **[C]**
- $a + 3a + 5a + 7a = 1$
 $16a = 1 \therefore a = \frac{1}{16}$
USING THIS, FIND $E(x) = 17 \frac{7}{8}$ **[B]**
- SOLVE $0 = \cos(2(x - \pi/6))$
 $x = \frac{11\pi}{12}$. (YOU CAN SUB IN EA. ANSWER) **[C]**
- $\int_0^4 f(x) dx = 10$
 $= \int_0^3 f(x) dx + R_2 = 10$
 $= \int_0^3 f(x) dx - 3 = 10$
 $\therefore \int_0^3 f(x) dx = 13$
 $\int_{-1}^3 f(x) dx = R_1 + \int_0^3 f(x) dx$
 $= -2 + 13$
 $= 11$ **[C]**
- STAT. PT AT $x=0$.
INCREASING FOR $x < 0$.
DECREASING FOR $x > 0$.
APPROACHING ASYMPTOTE AS
 $x \rightarrow -\infty \neq x \rightarrow \infty$. **[B]**

SECTION II: EXT. RESPONSE

- 9a. $a=9$ $\left\{ \begin{array}{l} 2963 = 9 + (n-1) \times 7 \\ d=7 \end{array} \right. \left\{ \begin{array}{l} 2963 = 2 + 7n \\ 2961 = 7n \\ n = 423 \end{array} \right.$
 $S_{423} = \frac{423}{2} (2 \times 9 + (422 \times 7))$
 $= 628\,578$.
- 10a. $y' = 2 \cos 2x$
- β . $u = e^x$ $\times$ $v = x^2 + 1$
 $u' = e^x$ $\times$ $v' = 2x$
 $f'(x) = e^x(x^2 + 1) + 2xe^x$
 $= e^x(x^2 + 2x + 1)$
- γ . $u = x$ $\times$ $v = \tan x$
 $u' = 1$ $\times$ $v' = \sec^2 x$
 $y' = \tan x + x \sec^2 x$
 $x = \pi/3$ $y' = \tan(\pi/3) + \pi/3 \sec^2(\pi/3)$
 $= \sqrt{3} + \pi/3 \frac{1}{\cos^2(\pi/3)}$
 $= \sqrt{3} + \frac{4\pi}{3}$
- 10a. α . $\int \frac{x^2 + 2x}{x} dx = \int \frac{x^2}{x} + \frac{2x}{x} dx$
 $= \int x + 2 dx$
 $= \frac{x^2}{2} + 2x + C$

β. $\int (3x-4)^8 dx = \frac{(3x-4)^9}{27} + C.$

bi.



• UPPER SEMI-CIRCLE, SHIFTED RIGHT 2 UNITS, RADIUS 2 UNITS.

ii. $\int_2^4 \sqrt{4-(x-2)^2} dx$
= AREA OF $\frac{1}{4}$ OF CIRCLE.

$\therefore \text{AREA} = \frac{1}{4} \times \pi \times 2^2$
= π UNITS²

ci. $\text{AREA} = \frac{1}{2} ab \sin C$
= $\frac{1}{2} \times 6 \times 6 \sin 30^\circ$
= 9 cm^2

ii. $\text{SECTOR} = \frac{1}{12} \times \pi \times 6^2$
= $3\pi \text{ cm}^2$
SHADED SEGMENT
= $3\pi - 9 \text{ cm}^2$.

11a. $9-4 \cdot 7 = 4 \cdot 3$, $9-3 \cdot 8 = 5 \cdot 2$
 $A = \frac{3}{2} (4 + 4 \cdot 2 + 2(4 \cdot 3 + 5 \cdot 2))$
= $40 \cdot 8 \text{ m}^2$

• IT'S EASIER TO FIND LENGTHS ON LAND FIRST.

bi. $\frac{d}{dx} (\cos x)^5$

= $5 \cdot (-\sin x) (\cos x)^4$
= $-5 \sin x \cos^4 x$

ii. $\int \sin x \cos^4 x dx$
= $-\frac{1}{5} \int -5 \sin x \cos^4 x dx$
= $-\frac{1}{5} (\cos^5 x) + C$

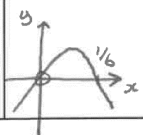
c. $f'(x) = 4x^3 - 3$
 $f(x) = x^4 - 3x + C$
 $f(-1) = (-1)^4 + 3 + C = 2$
 $4 + C = 2$
 $C = -2$

$\therefore f(x) = x^4 - 3x - 2$

12ai. $f(x) = 3x-1$, $g(x) = \frac{1}{2x}$
 $g(f(x)) = \frac{1}{2(3x-1)}$
= $\frac{1}{6x-2}$

ii. $D: x \in \mathbb{R}, x \neq \frac{2}{6}$
"ALL REAL x " $\neq \frac{1}{3}$ ☺

iii. $f'(x) = 2x - 12x^2$
INCREASING: $2x - 12x^2 > 0$
 $2x(1-6x) > 0$
 $\therefore 0 < x < \frac{1}{6}$
PART OF GRAPH ABOVE x -AXIS.

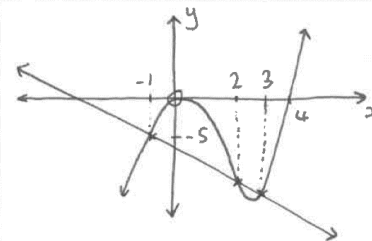


c. PT. OF INTERSECTION...

$x-5 = x^2-5x$
 $0 = x^2-6x+5$
 $0 = (x-5)(x-1)$
 $x = 5, 1$

$\text{AREA} = \int_1^5 (x-5) - (x^2-5x) dx$
= $\int_1^5 -x^2 + 6x - 5 dx$
= $\left[-\frac{x^3}{3} + 3x^2 - 5x \right]_1^5$
= $\left(-\frac{125}{3} + 75 - 25 \right) - \left(-\frac{1}{3} + 3 - 5 \right)$
= $\frac{32}{3}$ UNITS²

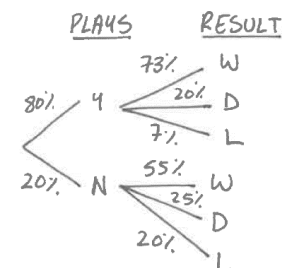
13ai



$f(x) = x^2(x-4)$
• x -INTS: 0, 4
• CUBIC: BOUNCE, CUT
• y -INT: 0

ii. $(-\infty, -1] \cup [2, 3]$

bi.



$P(\text{PLAY}, \text{WIN}) = 80\% \times 73\%$
= 58.4%

$P(\overline{\text{PLAY}}, \text{WIN}) = 20\% \times 55\%$
= 11%

$P(\text{WIN}) = 58.4\% + 11\%$
= 69.4%

ii. CONDITIONAL PROBABILITY:

$P(\overline{\text{LOSES}}) = 1 - P(\text{LOSES})$
= $1 - (80\% \times 7\% + 20\% \times 20\%)$
 $P(W, D) = 1 - 9.6\% = 90.4\%$

$P(\text{PLAYS} \cap \overline{\text{LOSES}}) = 58.4\% + (80\% \times 20\%)$
= 74.4%

PUTTING IT TOGETHER:

$P(\text{PLAYS} | \overline{\text{LOSES}})$
= $\frac{P(\text{PLAYS} \cap \overline{\text{LOSES}})}{P(\overline{\text{LOSES}})}$
= $\frac{74.4\%}{90.4\%} = 82.3\%$

- 14a. $P(x, y)$: DILATE $\frac{1}{3}$ HORIZ.
 ↓
 · TRANSL. +1 HORIZ.
 · DILATE $\frac{1}{2}$ VERT.
 ↓
 · TRANSL. +4 VERT.

$P'(x, y)$

· TO GET TO ORIGINAL, REVERSE THE OPERATIONS:

$P'(x, y)$: $(-6, 5)$
 ↓
 $(-6, 1)$ → -4 v.
 $(-6, 2)$ → × 2 v.
 ↓
 $(-7, 2)$ → -1 h.
 $P(x, y)$ $(-21, 2)$ → × 3 h.

OR

SUB. $P'(-6, 5)$ INTO TRANSFORMED FUNCTION:

$$5 = \frac{1}{2}g(3(-6-1)) + 4$$

$$1 = \frac{1}{2}g(3(-7))$$

$$2 = g(-21)$$

$$\therefore x = -21, y = 2.$$

bi. $T_1 = d$

$$T_2 = 0.85d + d$$

$$T_3 = 0.85(0.85d + d) + d$$

$$= 0.85^2d + 0.85d + d$$

$$T_n = 0.85^{n-1}d + \dots + 0.85d + d$$

THIS IS A GP WITH:

$$\left. \begin{array}{l} a = d \\ r = 0.85 \end{array} \right\} S_n = \frac{d(1-0.85^n)}{1-0.85}$$

$$= \frac{d(1-0.85^n)}{0.15}$$

$$= \frac{20d(1-0.85^n)}{3}$$

ii. $T_7 = \frac{20d(1-0.85^7)}{3} = 50$

$$d \times 4.529 \dots = 50$$

$$d = 11.038 \dots$$

$$= 11 \text{ mg.}$$

15ai. AMPLITUDE: $(16-10) \div 2 = 3$

PERIOD: $\frac{2\pi}{b} = 12$... -3 FOR FLIPPED COS.

IT'S COS GRAPH $b = \pi/6$ FROM GRAPH

ii. EQN: $h = -3 \cos(\pi/6 t) + 13$

$$11.5 = -3 \cos(\pi/6 t) + 13$$

$$-1.5 = -3 \cos(\pi/6 t)$$

$$0.5 = \cos(\pi/6 t)$$

$$\therefore \cos \phi = 0.5$$

Q1: $\phi = \pi/3$

$$t = \pi/3 \div \pi/6 = 2$$

Q4: $\phi = 5\pi/3$

$$t = 5\pi/3 \div \pi/6 = 10$$

$$8\text{AM} + 2 = 10\text{AM}$$

$$8\text{AM} + 10 = 6\text{PM}$$

OR

READ OFF THE GRAPH! (??)

CONT'D →

b. REWRITING:

$$-\log_2 x - 2\log_2 x - 3\log_2 x + \dots = -110$$

THIS IS AN AP WITH:

$$\left. \begin{array}{l} a = -\log_2 x \\ n = 10 \\ L = -10 \log_2 x \end{array} \right\}$$

$$S_{10} = \frac{10}{2} (-\log_2 x - 10 \log_2 x)$$

$$-110 = 5 (-11 \log_2 x)$$

$$-110 = -55 \log_2 x$$

$$2 = \log_2 x$$

$$\therefore x = 4.$$

16i. $A_1 = 650\,000 \times 1.005 - 4200$

$$A_2 = (650\,000 \times 1.005 - 4200) \times 1.005 - 4200$$

$$= 650\,000 \times 1.005^2 - 4200 \times 1.005 - 4200$$

$$= 650\,000 \times 1.005^2 - 4200(1.005 + 1)$$

$$A_3 = [650\,000 \times 1.005^2 - 4200(1.005 + 1)] \times 1.005 - 4200$$

$$= 650\,000 \times 1.005^3 - 4200(1.005^2 + 1.005 + 1)$$

YOU NEED TO SHOW MORE WORKING

$$A_n = 650\,000 \times 1.005^n - 4200(1.005^{n-1} + \dots + 1)$$

$$\left. \begin{array}{l} a = 1 \\ r = 1.005 \end{array} \right\} \left[\begin{array}{l} S_n = \frac{1(1.005^n - 1)}{0.005} \\ = 200(1.005^n - 1) \end{array} \right]$$

$$\therefore A_n = 650\,000 \times 1.005^n - 4200 \times 200(1.005^n - 1)$$

CONT'D →

$$= 650\,000 \times 1.005^n - 840\,000(1.005^n - 1)$$

$$= -190\,000 \times 1.005^n + 840\,000$$

EXPAND & GROUP LIKE TERMS.

ii. $A_n = \text{AMT OWING} \times 1.005^n - 840\,000(1.005^n - 1)$

· USING EQUATION FROM ABOVE.

· NOW, AMT OWING IS:

$$\$373\,722 + 200\,000$$

$$= \$573\,722$$

$$\therefore 0 = 573\,722 \times 1.005^n - 840\,000(1.005^n - 1)$$

↑
NOTHING OWING AT END.

$$0 = 840\,000 - 266\,278 \times 1.005^n$$

$$\frac{-840\,000}{-266\,278} = 1.005^n$$

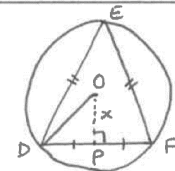
$$3.154 \dots = 1.005^n$$

$$n = \frac{\log_e 3.154 \dots}{\log_e 1.005}$$

$$= 230.3$$

$$= 231 \text{ MONTHS.}$$

17i.



$$\text{AREA} = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times DF \times (10 + x) \quad \dots \text{RADIUS} = 10$$

$$= DP \times (10 + x)$$

$$OD = 10 \text{ cm (RADIUS)}$$

$$DP^2 = 10^2 - x^2$$

$$DP = \sqrt{100 - x^2}$$

$$\therefore \text{AREA} = \sqrt{100 - x^2} \times (10 + x)$$

FINDING DP USING PYTHAG

ii. $A = (10+x)\sqrt{100-x^2}$
 $u = 10+x$ ~~$v = (100-x^2)^{1/2}$~~
 $u' = 1$ ~~$v' = -x(100-x^2)^{-1/2}$~~
 $= \frac{-x}{\sqrt{100-x^2}}$

$$\therefore \frac{dA}{dx} = \frac{\sqrt{100-x^2} - x(10+x)}{\sqrt{100-x^2}}$$

$$= \frac{100-x^2 - x(10+x)}{\sqrt{100-x^2}}$$

$$= \frac{100-x^2-10x-x^2}{\sqrt{100-x^2}}$$

$$= \frac{100-10x-2x^2}{\sqrt{100-x^2}}$$

STAT. PTS WHEN: $100-10x-2x^2=0$

$$-2(x^2+5x-50)=0$$

$$-2(x+10)(x-5)=0$$

$$\therefore x = -10, 5$$

$$\text{DENOM} = 0$$

TEST IF $x=5$ PRODUCES MAX.

USE 1ST DERIV. TEST:

x	4	5	6
$\frac{dA}{dx}$	$\frac{28}{\sqrt{84}}$	0	-4

$\therefore x=5$
IS A MAX.

MOST SHOW $\frac{dA}{dx}$ NUMBERS.

SHOW ALL SIDES ARE EQUAL.

$$DP = \sqrt{100-5^2} \quad \dots \text{FROM i.}$$

$$= \sqrt{75} = 5\sqrt{3}$$

$$DF = 2 \times DP$$

$$= 10\sqrt{3} \quad \dots \text{1ST SIDE}$$

$$DE^2 = (10+5)^2 + (\sqrt{75})^2$$

$$= 225 + 75$$

$$= 300$$

$$DE = \sqrt{300} = 10\sqrt{3} \quad \dots \text{2ND SIDE}$$

$$DF = DE = 10\sqrt{3} \quad \dots \text{3RD SIDE}$$

$\therefore$ EQUILATERAL TRIANGLE!

END! PHEW! 😊